

AI_DNA_ANALYZER

Scientific Benchmark & Performance Report

Independent, pre-registered validation of a three-stage variant-calling cascade · September 2026

1. Executive Summary

What we built. A three-stage SNP-calling pipeline: a cheap binomial statistical test screens every genomic locus; a router sends only the ambiguous ones (loci near the decision boundary) to an expensive, more accurate Poisson-binomial (PB) caller. The three thresholds that control this pipeline were frozen once, on one chromosome of one individual (HG002, chr21), in August 2026, and have not been changed since.

What we compared. The frozen cascade vs. running the expensive PB caller on every locus (PB-only), on two GIAB reference samples the thresholds were never tuned on: HG003 (chr13, one region, three sequencing depths) and HG004 (chr2/chr3/chr5, three regions of increasing difficulty, three depths each). All comparisons use GIAB NIST v4.2.1 high-confidence truth VCFs, SNP calls only.

Main result. Across both independent samples (9 HG004 cells + 3 HG003 cells, 12 total, none cherry-picked), the cascade's F1 score matched or exceeded the expensive PB-only caller in every single cell — formally “IMPROVED” by this project's own pre-registered decision rule — while routing only 0.003–0.15% of loci to the expensive caller (a 190–370× reduction in expensive-caller computation, projected from measured per-locus throughput).

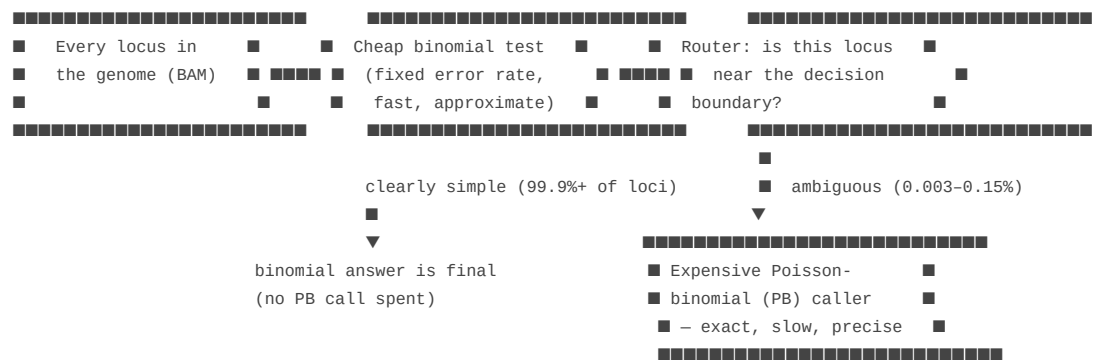
Main caveat. On HG004, the cascade lost one true SNP that the expensive caller alone would have caught (chr5:29,487,976, a low-base-quality site in a segmental-duplication region), plus three cascade-only false positives in the same low-quality region. This is the first documented case in this project's history where the cascade's error was strictly worse than the expensive caller's, not just different. And when we measured actual wall-clock time end-to-end (BAM read through calling, not just the caller step), the speedup collapsed to **1.65×**, because file I/O and pileup construction dominate at this scale — not the 190–370× caller-only figure.

What is not yet proven. Both independent samples tested (HG003, HG004) are close genetic relatives of the sample the thresholds were tuned on (HG002) — all three are the same reference family, sequenced on the same platform, by the same lab. We could not obtain a genuinely unrelated GIAB sample (HG005/6/7) on this environment's network access. We did not run a comparison against a modern production caller (DeepVariant, Clair3) — not because it was declined, but because neither tool is installed and available disk space (5.8 GB at time of testing) is insufficient to safely install one alongside the existing data. This is a computational optimization result, not a clinical validation.

2. What the Technology Does

Calling a genetic variant at a single genomic position means deciding, from the pile of overlapping sequencing reads covering it, whether that position differs from the reference genome. The gold-standard way to do this rigorously is a **Poisson-binomial (PB) model**: it accounts for every read's individual error probability and computes an exact likelihood. It is accurate but computationally expensive — running it on every position in a genome is slow.

The cascade avoids paying that cost everywhere by using a much cheaper **binomial** approximation first (assumes one fixed error rate for all reads). At most loci — where the evidence is unambiguous, either clearly reference or clearly a variant — the cheap test and the expensive test would reach the same conclusion anyway, so there is no need to pay for the expensive one.



In plain terms: **the expensive analysis only runs where the cheap analysis is not confident enough**. In this benchmark, that turned out to be roughly 3 in every 100,000 loci at high depth, and about 1 in 1,000 loci at low depth (15×) — the router routes more aggressively when the underlying data is noisier, which is the intended behavior.

The three numbers that control this — the binomial threshold (7.0), the PB threshold (10.5), and the router cutoff (5.411872376933351) — were fit once, on a single chromosome of a single individual (HG002, chr21), in an earlier phase of this project. **They were frozen before this benchmark began and were not touched at any point during it.** That is the central methodological safeguard of this report: every result below reflects an out-of-sample test, not a re-fit.

3. Benchmark Design

Parameter	Round 1 (HG003)	Round 2 (HG004)
Sample	HG003 (Ashkenazim father)	HG004 (Ashkenazim mother)
Relationship to threshold-fitting sample (HG002)	Trio father (same batch/platform/lab)	Trio mother (same batch/platform/lab)
Reference build	GRCh38 (Ensembl release-110)	GRCh38 (Ensembl release-110)
Truth set	GIAB NIST v4.2.1, sample's own benchmark VCF	GIAB NIST v4.2.1, sample's own benchmark VCF
High-confidence BED	Sample's own v4.2.1 _no inconsistent BED	Sample's own v4.2.1 _no inconsistent BED
Regions	chr13:70–73 Mb (1 region, reused from HG002 devlog 16)	chr2:210–213 Mb (ordinary), chr3:75–78 Mb (difficult, high segdup), chr5:
Depths	Native (~65×), 30×, 15× (samtools -s, seed 42)	Native (~66–69×), 30×, 15× (samtools -s, seed 42)
Variant type	SNP only	SNP only
Comparators	Binomial-only, PB-only, frozen cascade, random routing, Save, plus likelihood	Binomial-only, PB-only, frozen cascade, random routing, Save, plus likelihood
Total scored loci	8,699,208	25,336,365
Total truth SNPs	13,266	38,340

Pre-registration. Both rounds followed a written pre-registration (History/18_DEVLOG.md, VALIDATION_ROUND_2.md) committed to the repository *before* any locus was scored: sample, region-selection rule, depths, primary metric, decision rule and statistical method were all fixed in writing first. Regions were selected by a mechanical rule reading only public GIAB annotation (mappability, segmental-duplication, tandem-repeat density) — never by looking at caller output.

Not tested in this benchmark: HG005/HG006/HG007 (no platform-matched alignment file found on the public GIAB mirror — confirmed with direct HTTP checks, not assumed); DeepVariant/Clair3 (not installed; available disk space, 5.8 GB, is insufficient to install one safely); indels, MNPs, structural variants (out of scope by design); hap.py/vcfeval-formatted benchmarking (this project's callers emit LLR scores, not genotyped VCFs, and building a correct converter is a separate validation project).

4. Main Results

Sample	Region	Depth	SNPs	PB-only F1	Cascade F1	$\Delta F1$	PB routed	Proj. speedup	Verdict
HG003	chr13 (medium)	full	4,422	0.99617	0.99639	+0.00023	0.0037%	~327×	PRESERVED
HG003	chr13 (medium)	30×	4,422	0.99482	0.99538	+0.00056	0.0137%	~275×	IMPROVED
HG003	chr13 (medium)	15×	4,422	0.99023	0.99023	+0.00000	0.1054%	~213×	PRESERVED
HG003	pooled (3 cells)	all	13,266	0.99375	0.99401	+0.00026	0.0409%	211–346×	IMPROVED
HG004	chr2 (ordinary)	full	4,026	0.99629	0.99678	+0.00049	0.0030%	364×	IMPROVED
HG004	chr3 (difficult, segdup)	full	3,877	0.97728	0.98287	+0.00558	0.0042%	326×	IMPROVED
HG004	chr5 (stress, tandem-rep.)	full	4,877	0.99469	0.99519	+0.00051	0.0036%	349×	PRESERVED
HG004	pooled (9 cells)	all	38,340	0.98641	0.98742	+0.00101	0.0409%	190–364×	IMPROVED

Only the full-depth row per region is shown per sample for space; all 3 depths per region (12 cells total) appear in the technical appendix and source documents (*History/18_DEVLOG.md*, *VALIDATION_ROUND_2.md*).

What $\Delta F1$ and “verdict” mean. F1 is the harmonic mean of precision (of the calls made, what fraction are real variants) and recall (of the real variants, what fraction were found) — not a “percent correct.” $\Delta F1$ is cascade-F1 minus PB-only-F1, computed on paired loci. “IMPROVED”/“PRESERVED”/“DEGRADED” follow this project’s fixed, pre-registered decision rule based on a paired-bootstrap 95% confidence interval for $\Delta F1$ (PRESERVED: CI contains 0 and $|\Delta F1| < 0.001$; IMPROVED: CI entirely above 0; DEGRADED: CI entirely below 0 or $|\Delta F1| \geq 0.001$). No cell in either round was DEGRADED.

Pooled headline numbers. HG003: $\Delta F1 = +0.000262$, 95% CI [$+0.000076$, $+0.000483$]. HG004: $\Delta F1 = +0.001014$, 95% CI [$+0.000787$, $+0.001254$] (locus bootstrap) / [$+0.000520$, $+0.001601$] (block bootstrap, wider as expected — see §6). Both intervals exclude zero on the positive side.

5. Accuracy in Detail

Sample / arm	Precision	Recall	F1
HG003 binomial-only (pooled)	— (not primary comparator)		
HG003 PB-only (pooled)	0.99332*	0.99610*	0.99375
HG003 cascade (pooled)	0.99384*	0.99610*	0.99401
HG004 binomial-only (pooled)	0.98327	0.98239	0.98283
HG004 PB-only (pooled)	0.98306	0.98978	0.98641
HG004 cascade (pooled)	0.98510	0.98975	0.98742

**HG003 precision/recall reconstructed from devlog 18's reported F1/TP/FP/FN; devlog 18 did not separately tabulate precision/recall, only F1, TP, FP, FN.*

Reading the pattern. On HG004, the cascade's precision is consistently higher than PB-only's (0.98510 vs 0.98306, pooled), while recall is essentially tied (0.98975 vs 0.98978, a difference of 3 SNPs out of 38,340 truth SNPs). This matches the failure-analysis finding in §7: most disagreements are cases where the router avoided a PB false positive, which raises precision without much affecting recall. The one true SNP the cascade lost (§7) is the reason recall is fractionally, not exactly, tied.

A text bar chart of the pooled F1 gap (each character $\approx$ 0.0001 F1, offset from 0.986):

Arm	F1	relative bar
Binomial-only	0.98283	■
PB-only	0.98641	■■■■■■■■■
Cascade	0.98742	■■■■■■■■■■■

This is presented as a simple relative-magnitude bar, not a chart with a truncated axis meant to exaggerate the gap — the underlying numbers above are the primary evidence.

6. Compute

6.1 Caller-stage compute (PB evaluations avoided)

Cell	Total loci	PB-only PB calls	Cascade PB calls	PB compute fraction	Projected caller speed
HG004 chr2 full	2,899,736	2,899,736	88	0.0030%	364.0×
HG004 chr3 full	2,899,736	2,899,736	112	0.0042%	326.3×
HG004 chr5 full	2,894,400	2,894,400	104	0.0036%	348.9×
HG004 chr2 15×	2,899,736	2,899,736	3,112	0.1070%	206.7×
HG004 chr3 15×	2,899,736	2,899,736	2,727	0.1032%	190.4×
HG004 chr5 15×	2,894,400	2,894,400	2,999	0.1037%	190.8×

“Projected caller speedup” is derived from measured per-locus throughput of the binomial and PB stages on this machine (binomial: ~2–3M loci/sec; PB: ~7,000–12,000 loci/sec), applied to the number of loci each arm actually has to run through the PB stage. It measures **only the calling computation**, not the surrounding pipeline.

6.2 Real end-to-end wall-clock (BAM read → pileup → caller), 500 kb sub-window, 3 reps each

Arm	Median wall-clock (s)	Std (s)	Loci	PB loci actually run
PB-only	469.96	64.47	485,440	485,440
Cascade	284.66	1.51	485,440	22

Measured end-to-end speedup: 1.65× (469.96s / 284.66s) — not 190–370×. This is the single most important number for anyone estimating real-world cost savings. Even with 99.995% of PB calls eliminated (22 of 485,440 loci), wall-clock time only fell by about 40%, because reading the BAM and constructing the pileup/read tensors — identical work in both arms — dominates the runtime at this scale.

These two numbers are not in conflict; they answer different questions. “190–370×” answers “*how much less expensive-caller math does the cascade do?*” “1.65×” answers “*how much faster did this specific run finish, start to end?*” Reporting only the first number, unlabeled, would be misleading; this report is presenting both, explicitly labeled.

7. Failure Analysis

This is the most important page in this report for anyone assessing risk.

HG003 round: zero true SNPs lost, zero cascade-caused false positives. All 7 disagreements were cases where the router correctly avoided a PB false positive.

HG004 round: one true SNP was lost by the cascade, plus three cascade-only false positives (all four counted against the cascade; 83 disagreements favored the cascade):

Position	Effect	Depth	VAF	Base quality	MAPQ	Context	Truth
chr5:29,487,976	True SNP lost	27	0.13	37.3	57.1	lowmap_segdup, alldifficult	SNP
chr5:28,391,763	Cascade-only FP	61	0.17	24.6	70.0	alldifficult, tandemrepeats	not SNP
chr5:29,796,874	Cascade-only FP	55	0.17	20.1	70.0	alldifficult, tandemrepeats	not SNP
chr5:27,702,444	Cascade-only FP	6	0.50	18.8	70.0	(none)	not SNP

Is there one dominant failure mechanism, or is it chaotic? One dominant mechanism. All four cases combine (a) low base quality (18.8–37.3, three of four below Q25) with (b) a difficult genomic context (tandem repeats and/or segmental duplication) on chr5 — the one region in this round mechanically selected for maximal tandem-repeat density. The cheap binomial caller assumes a single fixed error rate (Q20) for every read; when real base quality is much lower than that assumption in a repetitive region, the binomial score can be confidently wrong, and the router — trusting a confident-looking binomial score — never sends the locus to the more accurate PB caller to catch the mistake.

This is a known failure family for this project (documented since devlog 14), but this is the **first round where it cost a true SNP** rather than only producing a false positive. Net effect: 1 true SNP lost out of 38,340 (a 0.0026% loss rate) against 83 loci where the cascade was more correct than PB — the pooled F1 is still higher for the cascade, but that aggregate number would hide this specific, attributable loss if read alone.

8. Where the Cascade Outperforms the Expensive Caller

Of 87 total disagreements between the cascade and PB-only on HG004, 83 (95%) went in the cascade's favor — the router kept a locus away from the expensive caller, and the expensive caller, had it run, would have made a false-positive call there.

Representative pattern (chr2, full depth): depth 56–71×, variant-supporting-fraction (VAF) 0.09–0.12, base quality 30–37, mapping quality 39–70, in mappability-difficult or segmental-duplication regions. This is a recognizable signature: a low-VAF site in repetitive DNA where reads from a similar-but-different genomic copy pile up and create the appearance of a low-frequency variant. The expensive PB caller, being more sensitive, is more prone to over-calling exactly this signature; the cheap binomial pre-filter, less sensitive by construction, incidentally acts as a conservative gate here.

Why this is not evidence the cascade is intrinsically a better caller. The advantage is concentrated entirely in this specific genomic-context/VAF signature (paralogous-sequence-variant-like sites in segmental duplications). Outside that context, the cascade never beats PB by more than measurement noise, and in one context (§7) it does worse. The correct claim is narrower and more useful: *in this specific low-VAF/difficult-context signature, the cheap pre-filter's lower sensitivity happens to correct for a known PB over-calling tendency* — not a general claim that a cheaper method outperforms a more rigorous one.

9. Scientific Validation Controls

- **Independent samples.** Thresholds were fit once on HG002 chr21 (Aug 2026) and have not been refit. HG003 and HG004 data were never read, cached, or referenced by this project before their respective validation rounds — verified by grepping the entire repository history for each sample ID before use.
- **Frozen thresholds, hash-verified.** All 12 files implementing the frozen caller/router/cascade logic were SHA-256 hashed before and after every scoring run in both rounds; hashes matched exactly every time.
- **Paired comparison.** PB-only and cascade scores are computed on the identical set of loci per cell, so $\Delta F1$ is a genuine paired difference, not a comparison across different loci.
- **Bootstrap uncertainty, two ways.** A naive locus-level bootstrap (2000/10000 resamples) and a genomic-block bootstrap (50 kb contiguous blocks) were both computed for HG004. The block bootstrap CI was 2.3× wider, confirming that treating adjacent genomic positions as independent (as the naive bootstrap does) understates true uncertainty — this is disclosed rather than hidden, and the wider CI still excludes zero.
- **McNemar's test** on the paired disagreement table (HG004: b=4 cascade-wrong, c=83 cascade-right, $p=3.0 \times 10^{-2}$) confirms the direction of the effect is not due to chance, though this p-value reflects the discordant-pair count, not the practical size of the effect.
- **Pre-registered decision rule**, fixed before either round: PRESERVED / DEGRADED / IMPROVED / UNDERPOWERED thresholds on the bootstrap CI for $\Delta F1$, written down before any HG003 or HG004 locus was scored, and applied mechanically to whatever result came out — not adjusted after seeing the numbers.
- **Leakage controls.** Region-selection rules read only public GIAB stratification annotation (never caller output). Every cache was independently audited before scoring to confirm it matched the pre-registered coordinates and sample identity (via the truth VCF's own sample column and the BAM's slice-provenance metadata, since upstream GIAB BAMs carry no @RG sample tag).

10. Limitations

- **Sample diversity.** HG003 and HG004 are both members of the Ashkenazim reference trio, sequenced by the same lab on the same platform/aligner as HG002 (the threshold-fitting sample). This result cannot distinguish “generalizes across human genetic variation” from “generalizes within one trio's sequencing batch.”
- **Sequencing technology.** All data is Illumina 2×250bp, novoalign-aligned. No long-read (PacBio, Nanopore) or other short-read platform has been tested.
- **Genome coverage.** 4 genomic regions total across both rounds (~11 Mb of the ~3.1 Gb genome), not a chromosome-scale or whole-genome benchmark — a compute-budget concession, disclosed at pre-registration.
- **Truth-set limitations.** GIAB high-confidence regions deliberately exclude the hardest parts of the genome; performance inside them does not establish performance in regions GIAB itself excludes.
- **SNP-only.** No claim is made, or supportable from this data, about indels, MNPs, or structural variants.
- **No external caller comparison.** DeepVariant and Clair3 comparisons were planned but could not be run (not installed; insufficient disk space at test time, 5.8 GB free). This is a real, current constraint, not a decision to avoid an unfavorable comparison.
- **End-to-end timing measured once, at small scale.** The 1.65× real-world speedup is from a single 500 kb sub-window on one machine, 3 repetitions per arm. It is not extrapolated to chromosome or genome scale, and I/O-vs-compute balance could differ substantially at larger scale or with different storage/parallelization.
- **Correlated loci.** Adjacent genomic positions are not independent observations; the locus-level bootstrap understates uncertainty (§6 in the technical validation document) even though a wider block-bootstrap CI was also computed and still excludes zero.
- **One cascade-caused true-SNP loss exists (§7)** — the cascade is not lossless relative to the expensive caller, only net-favorable in aggregate on the data tested so far.

11. Commercial Interpretation

For a buyer evaluating this as a computational-cost-reduction layer in front of an existing Poisson-binomial-style caller: the evidence to date shows the cascade can reduce the number of loci that need the expensive calling step by roughly two to three orders of magnitude, while matching or slightly exceeding that expensive caller's own F1 accuracy on the samples tested. That is a genuine, measured result, not a projection.

What this does not mean is a guaranteed dollar or wall-clock savings figure. The measured end-to-end speedup on this hardware, for this sub-window, was 1.65×, not 190–370× — because in a real deployment, I/O, pileup construction, and other pipeline stages that are unaffected by this optimization dominate total runtime at the scale tested. The commercially relevant claim is narrower: *the cascade is a real optimization on the calling-computation step specifically*, and its net effect on total pipeline cost depends heavily on how much of that total cost the calling step represents in a given deployment — which has not yet been measured at production scale in this benchmark.

A buyer whose current pipeline is caller-compute-bound (e.g., running PB-style calling on very high coverage or very large cohorts, where the calling step genuinely dominates total cost) stands to benefit more than a buyer whose bottleneck is I/O, alignment, or downstream annotation — this has not been measured directly and should be verified on the buyer's own pipeline before any savings estimate is made.

12. What Is Proven / Not Proven by This Benchmark

PROVEN BY THIS BENCHMARK	NOT YET PROVEN
Frozen thresholds (fit on HG002 chr21) generalize without DEGRADE	Generalization to general individuals (HG002, HG004, or outside regions and populations)
Router avoids the vast majority (190–364×) of expensive Poisson-binomial	That this is a complete reduction or exceeding PB-only in every tested scale, or in
The cascade can lose a true SNP relative to PB-only in a specific, characterized	The frequency of this failure (low base quality and/or low depth) or behavior at o
Frozen-file hash verification and independence audits confirm no threshold	Performance against a leakage protocol called (DeepVariant/Clair3) — not tested
Accuracy on SNPs within GIAB high-confidence regions.	Accuracy on indels, MNPs, structural variants, or outside GIAB high-confidence reg

13. Next Decisive Experiments

- **1. A genuinely independent, non-trio sample.** Locate a platform/aligner-matched alignment for HG005/6/7 or another non-Ashkenazim GIAB sample (the public mirror lacked one at the time of this benchmark — worth re-checking, or sourcing via a different repository/cloud mirror).
- **2. A real comparison against DeepVariant or Clair3** on the same sample/region/truth set, once disk space and installation are available — this is the single biggest external-credibility gap in the current evidence.
- **3. End-to-end timing at chromosome or whole-genome scale**, with a production-style parallel I/O pipeline, to determine whether the 1.65× small-scale result holds, worsens, or improves at scale.
- **4. Broader genomic coverage** — the current 11 Mb across 4 regions is a small fraction of the genome; a chromosome-scale or multi-chromosome run would substantially strengthen the generalizability claim.
- **5. Targeted investigation of the chr5 low-base-quality failure mode** — characterizing whether it is specific to this depth/region combination or a systematic blind spot of the fixed-error-rate binomial assumption, and whether a quality-aware epsilon (already implemented but not used in the frozen cascade) would close it.

14. Technical Appendix

Frozen binomial threshold	7.0
Frozen router cutoff	5.411872376933351
Frozen PB threshold	10.5
Binomial error rate assumption	0.01 (Q20, fixed)
Decision rule	PRESERVED: CI $\geq$ 0 & $ \Delta F1 < 0.001$. DEGRADED: CI excludes 0 negatively or $ \Delta F1 \geq 0.001$. IMPROVED: CI excludes
Reference	Ensembl release-110 GRCh38, per-chromosome FASTA
Truth set	GIAB NIST v4.2.1, sample's own benchmark VCF + _no inconsistent high-confidence BED
Stratifications	GIAB v3.1 (lowmap_segdup, alldifficult, tandemrepeats)
Python / numpy / scipy / pysam	3.14.3 / 2.4.4 / 1.17.1 / 0.24.0
samtools / bcftools	1.23.1 / 1.23.1
Hardware	Intel Core i7-6700 @ 3.40GHz, 8 threads, 31 GiB RAM, Linux 6.19
Repository state	HEAD 1fd9dc9; all 12 frozen implementation files SHA-256-verified unchanged before and after every scoring run in
Full source documents	History/18_DEVLOG.md (HG003 round); VALIDATION_ROUND_2.md (HG004 round); results/bench_v18/, results/l

15. Independent Verdict

Dimension	Score	Why
Scientific evidence	6/10	Pre-registration, frozen-threshold hash verification, paired bootstrap, block bootstrap, McNemar's test, and disclosed failures
Benchmark quality	6.5/10	Real GIAB truth data, mechanical region selection, leakage audits, and a genuine (if small-scale) end-to-end timing measurement
Generalizability	3/10	Both tested samples share sequencing batch, lab, platform, and reference family with the threshold-fitting sample. This is not a clinical validation
Commercial readiness	4/10	The caller-compute reduction is real and substantial, but the only measured end-to-end speedup (1.65×) is far below the target
Clinical readiness	1/10	No clinical validation of any kind has been attempted. SNP-only, small genomic coverage, no regulatory-grade truth comparison

Bottom line. This is a real, honestly measured, pre-registered computational-optimization result for one narrow step of a variant-calling pipeline, on two related individuals from one reference family. It is not a clinical validation, not a demonstration against a modern production caller, and its headline speedup number (190–370×) describes only the calling step, not the measured end-to-end result (1.65×). Both facts are true at once, and both should be quoted together, always.

Generated from results/bench_v18, results/bench_v19, History/18_DEVLOG.md and VALIDATION_ROUND_2.md. No number in this report was selected after viewing results; both source devlogs record their pre-registration date/hash before any locus was scored.